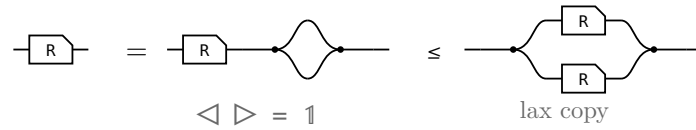


The allegory axioms: the proofs

The companion to `allegory-axioms.typ`, which states these laws and their pictures. Here each one is worked: the steps of the Lean proof, side by side, with the rule that reaches each under it.

§1. $R \circ R = R$, the one that is not bookkeeping



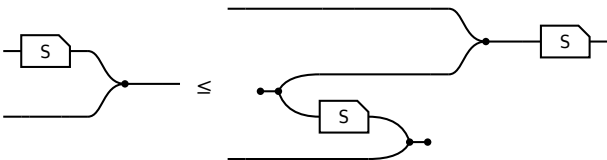
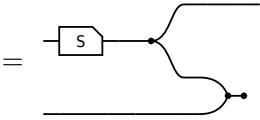
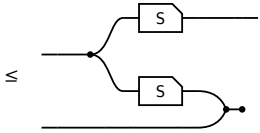
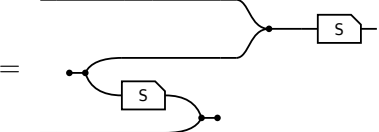
The last picture is $R \circ R$ itself, so this is $R \leq R \circ R$ and the lax copy law is the whole of it. The other direction is not proved here: $R \circ S \leq R$ holds for every S , by weakening S to $\top$ and then collapsing $R \circ \top$ — which is the paper’s own remaining three steps, packaged once.

§2. The modular law, from Frobenius

the modular law, $R \circ S \circ T \leq (R \circ T \circ S^\circ) \circ S$		
term	picture	the rule that reaches it
$(R \circ S) \circ T$		the start: $R \circ S \circ T$.
$= (\triangleleft (R \circ T)) ((S \circ 1) \triangleright)$		Factor out the shared prefix: $(R \circ S) \circ T$ is $(R \circ T) \circ (S \circ 1)$.
$\leq (\triangleleft (R \circ T)) ((1 \circ S^\circ) \triangleright S)$		The merge slide. S slides through the merge and reappears as S° on the other strand. The only inequality in the proof.
$= (R \circ (T \circ S^\circ)) \circ S$		Fold the prefix back in; the copy-merge pair reads as a meet again.

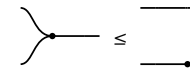
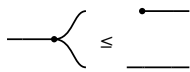
S occurs once on the left of the slide and twice on the right, and the lax copy law is the only law in the definition that may duplicate a box — so that is where the inequality has to be.

the merge slide, $(S \circ 1) \triangleright \leq (1 \circ S^\circ) \triangleright S$		
term	picture	the rule that reaches it
$(S \circ 1) \triangleright$		the start: $(S \circ 1) \triangleright$.

the merge slide, $(S \otimes 1) \triangleright \leq (1 \otimes S^\circ) \triangleright S$ 		
term	picture	the rule that reaches it
$= ((S \triangleleft) \otimes 1) (1 \otimes \triangleright \circ)$		Rebuild the merge from a copy and a right bracket, the shape the lax copy law applies to.
$\leq ((\triangleleft (S \otimes S)) \otimes 1) (1 \otimes \triangleright \circ)$		The lax copy law: $S \triangleleft$ becomes $\triangleleft (S \otimes S)$, the box copied. The whole of the modular law, in one step.
$= (1 \otimes S^\circ) (\triangleright S)$		Reshape back , and the surviving duplicate slides round the right bracket to become S° .

§3. $R \leq R R^\circ R$, by cutting two wires

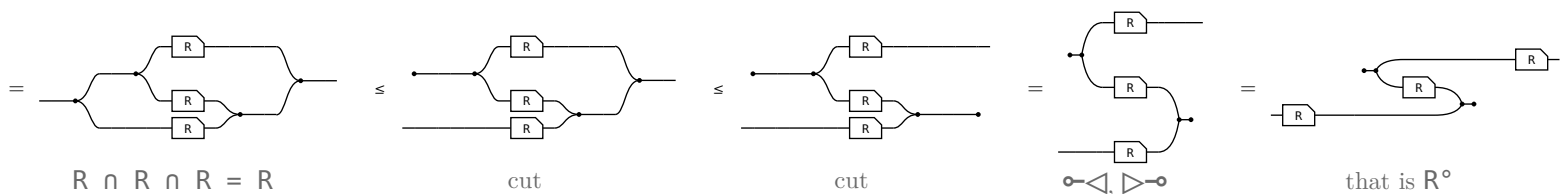
Freyd reads this off the modular law (§2.112): $R \sqsubseteq 1R \sqcap R \sqsubseteq (1 \sqcap R R^\circ) R \sqsubseteq R R^\circ R$. In pictures neither the modular law nor a meet with 1 is wanted. The one law that makes a picture bigger is spent twice, and it is the cut, $1 \leq \circ \circ$:



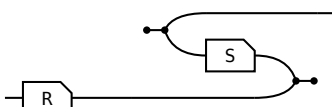
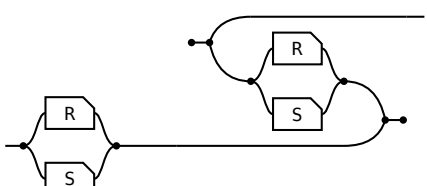
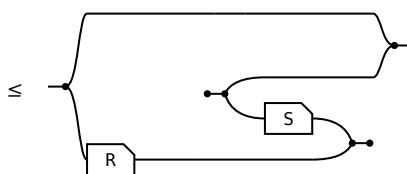
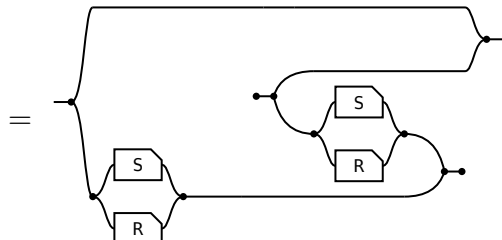
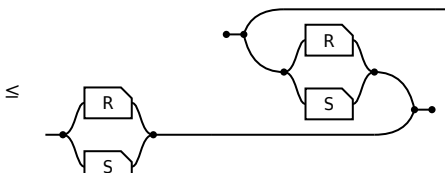
cut the copy's left output and what comes out there is **anything at all**

cut the merge's right input and it is **simply discarded**

R is its own three-fold meet — three copies of the box between a copy tree and a merge tree — and the two cuts turn the trees into a $\circ \triangleleft$ and a $\triangleright \circ$. The three boxes never move:

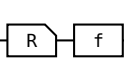
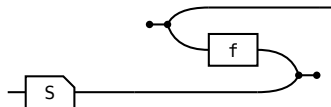
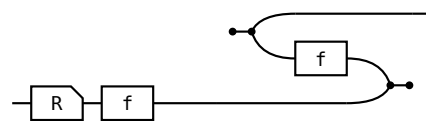
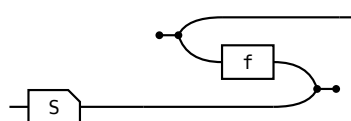


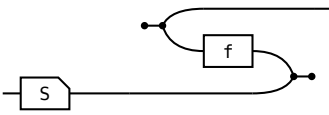
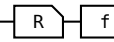
Read the last picture from the input, which is now the bottom strand: through the bottom box, up the $\triangleright \circ$ and back along the middle box — which is that box turned round, so it is R° — down the $\circ \triangleleft$ and out along the top box. Where the meet had one input feeding all three boxes and one output draining all three, the zig-zag has both of those **cut**, and a cut wire is the one thing this calculus lets you add for free.

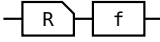
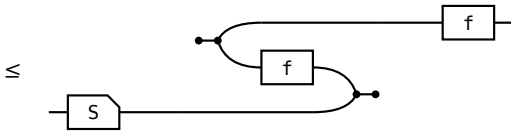
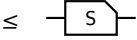
$\Leftarrow$ given $\text{---} \leq$  show $R \cap S$ entire, $\text{---} \leq$ 		
term	picture	the rule that reaches it
1	---	the start: 1.
$\leq 1 \cap (R S^\circ)$	$\leq$ 	the hypothesis, met with 1.
$= 1 \cap ((S \cap R) (S \cap R)^\circ)$	$=$ 	$1 \cap R S^\circ = 1 \cap (S \cap R) (S \cap R)^\circ$. $R S^\circ$ relates a to a' when some b has $a R b$ and $a' S b$; the $1 \cap$ sets $a' = a$, and it reads “some b has $a R b$ and $a S b$ ” — one arrow of $R \cap S$. Two arrows become one, and that is what needs the modular law.
$\leq (R \cap S) (R \cap S)^\circ$	$\leq$ 	$X \cap Y \leq Y$, then $S \cap R = R \cap S$.

§6. The shunting rule

A map moves from one side of a containment to the other at the cost of its converse, $R f \leq S \iff R \leq S f^\circ$. It is how a function gets out of the way so the relation underneath can be reasoned about. Each direction spends exactly one half of **map** — never both.

shunting right, $\Rightarrow$ given $\text{---} \text{---} \leq \text{---}$  show $\text{---} \leq$ 		
term	picture	the rule that reaches it
R		the start: R.
$\leq R (f f^\circ)$	$\leq$ 	total: $1 \leq f f^\circ$, so $f f^\circ$ may be inserted after R.
$\leq S f^\circ$	$\leq$ 	Re-bracket to $(R f) f^\circ$ and apply the hypothesis.

shunting right, $\Leftarrow$ given  $\leq$  show  $\leq$ 

term	picture	the rule that reaches it
$R \ f$		the start: $R \ f$.
$\leq S \ (f^\circ \ f)$		Apply the hypothesis on the left; the two fs are now adjacent.
$\leq S$		single valued: $f^\circ \ f \leq 1$, so the pair cancels.

Shunting left is the mirror, same shape, map on the other side throughout. What the rule is, is an adjunction: $f \dashv f^\circ$, with **total** the unit and **single valued** the counit, so “f is a map” and “f has a right adjoint, namely f° ” are one statement.